Initial Ideas

1. Basketball shot-tracking app
    - Keeps track of makes and misses, maintains count of makes to miss ratio as you continue
    - Can recognize person, basketball, and basket
    - Potential addition: If multiple people are shooting, then it can recognize individual shooting and keeps separate tracks of the two different people shooting
2. Calendar coordination app
    - Helps find time in your schedule to meet at convenient time/place
3. AI Coach/trainer/teacher (temp names: Tr**AI**n, Chiron)
    - Gives you a personalized plan and works with your unique prior knowledge/experience
    - Helps improve any "skill" you want to improve (ex. Sport, instrument, fitness, wellness, public speaking, school subjects, etc.) (emphasizes teaching/learning/growth)
    - Learns from your habits (if lesson goes bad/skipped a solution (new time, new exercise, etc. can be proposed)
    - Optimizes learning for you
    - Uses camera/microphone to watch/listen to your practice to help you improve (for skills)
        - <u>Features</u>:
        - <u>Calendar coordination</u>: Helps find time in your schedule to meet at convenient time/place
        - <u>Sports performance</u> (shot-tracking, form, etc.) <u>Tracking</u>
        - Keeps track of makes, misses, reps, etc. (ex: maintains count of makes to miss ratio as you continue)
        - Broken up into individual tutors for specific skills i.e. a guitar tutor, basketball coach, stats homework tutor

4. Social media "cleaner"
    - AI bot that fixes your feed to what you personalize it to instead of the social media algorithm that prioritizes engagement

Narrowed Idea (suggestions from Prof. Ren)

1. I suggest narrowing the scope of "skill" down to the **extreme**. Pick a very specific skill, something you are familiar with and you love.

Proposal (Ariel): For the AI coach, it takes a lot of data to train it and for instrument, public speaking learning, and shot-tracking it would be more difficult to train because it needs video/audio to interpret, then be corrected. Calendar coordination is an easy add on, but making a certain lesson difficulty customized to a specific user is a challenge. Tracking days an user participated in an activity and how long they've spent is easy to add, however.

2. Explain who your users are and what is the value for them to learn this skill.

Target users should be people getting introduced to training or sports, need some sort of guidance/starting point.

3. Use GPT deep research to identify all existing competitors for this particular thing and the market gap.

(For basketball shot-tracking app) there is an app that proposes this concept, but there is a paywall behind the app, and it is only available to iOS users. For a different app, you cannot live record a session, which limits to uploading video after. By making this accessible without a paywall and enabling both video upload and live session shots, the app we would create fills the gaps between the other competitors.

Potential Logo/name?

Narrowed stats based ideas:

- RPG build planner (focused on one game for this project): AI build planner for a single game which takes item and stat data to center a character build around
    - Inputs include what general build you want to follow, specifics like what weapon, armor, item, etc you want to use or stat you want to focus
    - Program should understand what items are compatible and what items aren't, i.e. if two items offer a similar bonus, but do not work in tandem, the program should not recommend using both and should instead offer a setup specializing in one or the other based on the player-provided information
    - Note: complexities of games will vary and will affect how in-depth the program needs to be to give accurate results
    - The target group for this program are people who want to plan out a build in advance for games that offer a large variety of tools to use in their completion. This encompasses new players who are unfamiliar with the depth of the game they are playing and would like advice on how to complete their experience with as few build planning troubles as possible, as well as veterans who feel they have run out of ideas on how to approach a game differently.
    - This program should also give notes on where the specified build may fall behind, either with specific encounters, enemies, classes, or effects.
    - Optionally: this program may be used as a guide mid-game against specific events or encounters, and should avoid spoiling important information regarding the game's plot or narrative if doing so has no bearing on the original purpose of providing a build. This means no statements like: "After [X Major Plot Event], do this" unless specified by the player that they do not need spoilers hidden.
    - For games that offer narrative choices and character alignments, the program should be capable of factoring in whether an action necessary to receive a component of the build requires "breaking role". In other words, if a good character would have to perform an evil action because the reward fit their build, then the program should at the very least warn the player and check for alternative items or routes.
    - Competition appears to be very limited, as most search results link to AI programs to develop your own RPG system or campaign for tabletop RPGs. Anything that falls under a build planner for RPGs are not AI programs that respond to user specifications, and are simply catalogues of equipment to apply to a character and require the user to already be very familiar with the build they want.

FINAL IDEA

Ethical stock correlation calculator/assistant: AI assistant that uses historical stock market data and trends to find more high integrity stocks that still yield positive returns.

- Stocks will have a social credibility score based on social perception, company goals/values, disclosed government/private relationships, historical accounts/past controversies, etc.
- High performing stocks with low social credibility scores will have different options for stocks with higher social credit that have high correlation with the high performing stock. (This can work across different markets)
- Helps the consumer feel positive about how their money is being used while invested. And offers an alternative to investing in megacorporations and general funds (some of which become heavily invested into singular stocks/markets)
- Offers options of high performing, ethical stocks across different markets to help the customer build a high performing, ethical, diversified portfolio based on the individual values of the customer
- Values can include things like: Climate awareness, worker treatment, political alignment (or potential neutrality), etc. (include future risk factors)
- Many competitors in the space offer premade ethical portfolios and funds, but this idea allows for a customizable portfolio based off of your specific values and what you want your money to support
- Stock market data, company history, market trends, etc. is all publicly available data
- The target customer is any long-term investor that wants to ethically grow their money via investments.
- Optionally: It may also be able to teach about proper investment habits (regular interval investing (not buying all at once due to emotion or drastic quick changes in the market, but calculated regular intervals))
- Monitor how individuals stand by their values with their money
- What will happen if nothing like this develops? (Bad future)
- Provide transparency (take action on your own)
- What documents can we get that are tough to understand, can we make them more readable for people? The news?
- Confidence score on how reliable our information is
- MVP: runs 24/7 keeps getting news/looks for legal document, organizes the information to be more understandable for the consumer

Heilmeier Catechism

1. What are you trying to do? Articulate your objectives using absolutely no jargon.

We are constructing an AI assistant that uses historical stock market data and trends to find more high integrity stocks that still yield positive returns. The AI will provide a social credibility score for stocks based on social perception, company goals/values, disclosed government/private relationships, and historical accounts, including past controversies.

2. How is it done today, and what are the limits of current practice?

How It Is Done Today: Directing an AI assistant to find companies on public trading stock market platforms, then using AI to recover news sources, company statements, and general data found online about the company to summarize the amount of information that the individual company matches with one's personal interests, then based on the gaps of information found, the AI can provide a confidence score to the investor, providing an explanation of whether the demands of the user are met or not. In addition, general stock market trends will be studied by the AI to determine whether the company investigated is worth investing in for making a profit.

Current limits: Limited amount of information regarding certain practices of some companies, possibility of decreased profit margin if money is invested in smaller companies.

3. What is new in your approach and why do you think it will be successful?

A major difference between this approach and others is the correlation-based stock replacement concept, which has not been done before. This approach allows investors to invest in markets they have interest in but choose a different type of company that better aligns with their values. Additionally, the confidence score determining whether the research ran by the AI contains sufficient information to make a definitive decision is a transparent, unique take that can lead the company to success, given that there is often a lack of conversation about the confidence that AI has over certain responses it provides.

4. Who cares? If you are successful, what difference will it make?

This project can provide people a greater sense of clarity and control over the actions they take by choosing to invest their money into companies that can provide both financial compensation and personal fulfillment. In addition, potential improvements from large-scale companies may occur in terms of consumer transparency and ethical motives coming from investors' pressure.

5. What are the risks?

One of the potential risks includes the AI assistant choosing companies that are not profitable or potentially cause the consumer to lose money rather than making a profit. Another risk is the AI

being incapable of finding sufficient information to offer a better alternative or providing incorrect information to an investor.

6. How much will it cost?

(Estimates taken from ChatGPT)

Monthly Costs for small-scale MVP

Infrastructure (~$1k): Cloud servers, databases, and hosting needed to run the backend, store financial data, and deliver the application to users.

Data feeds (~$300): APIs or licensed datasets that provide historical and current stock market prices, company fundamentals, and other financial metrics used for analysis.

AI processing (~$200): Computing resources used by machine learning or NLP models to analyze text, detect controversies, summarize reports, and generate credibility scores.

News/ESG scraping (~$200): Tools or services that collect public information such as news articles, sustainability reports, NGO data, and regulatory records used to evaluate companies' ethical impact.

7. How long will it take?

(Estimates taken from ChatGPT)

Initial prototype (6-8 weeks) to build the core system that takes into consideration stock data, calculates correlations, and generates credibility scores for each potential purchase.

MVP Product (8-10 weeks): This part of the process includes adding a user interface, valuing individual investor preferences, creating a portfolio builder for individual investors, and create a constructive scoring method.

Public launch (4-6 weeks): Train AI to improve scoring accuracy, expand coverage of stocks, fix potential bugs through testing finished product, include scaling and compliance.

8. What are the mid-term and final "exams" to check for success?

Mid-Term Exams

1. Technical feasibility: AI is capable of consistently finding alternative solutions correlated to original stocks.
2. Credibility score accuracy: AI is capable of reasonably scoring decisions it makes so it can provide transparency to investors while scores reflect real-world behavior.
3. Initial small-scale testing: Small batch of users provide feedback on scoring system and portfolios they have built and give scoring system approval/disapproval.

Final Exams

1. User Adoption/Retention: A large percentage of users are creating stock portfolios and using the platform regularly, and users choose to keep using the platform regularly over the span of several months.
2. Performance validation: Over long term testing, users find portfolios to perform reasonably close to traditional portfolios using large-scale or unethical companies.